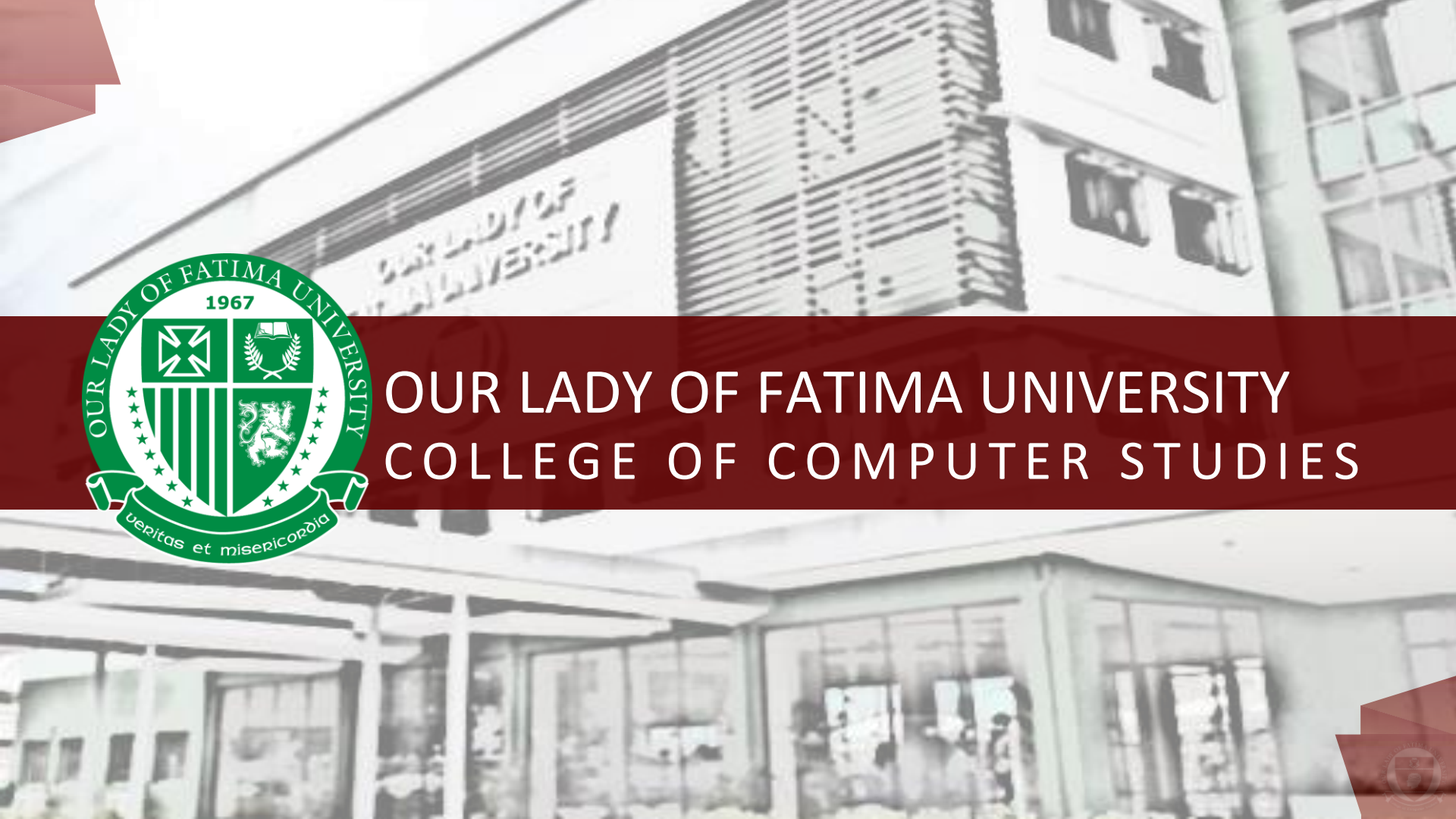
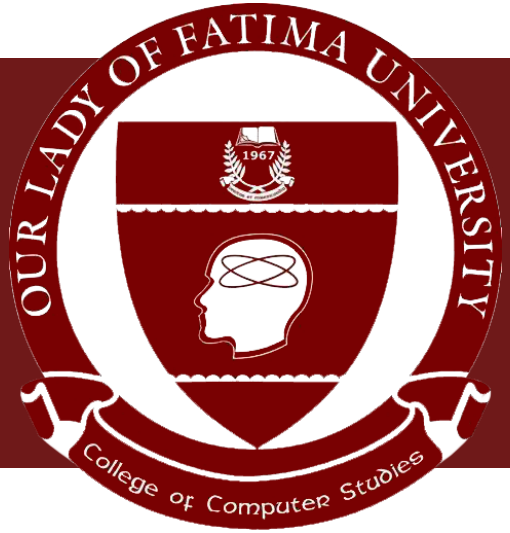
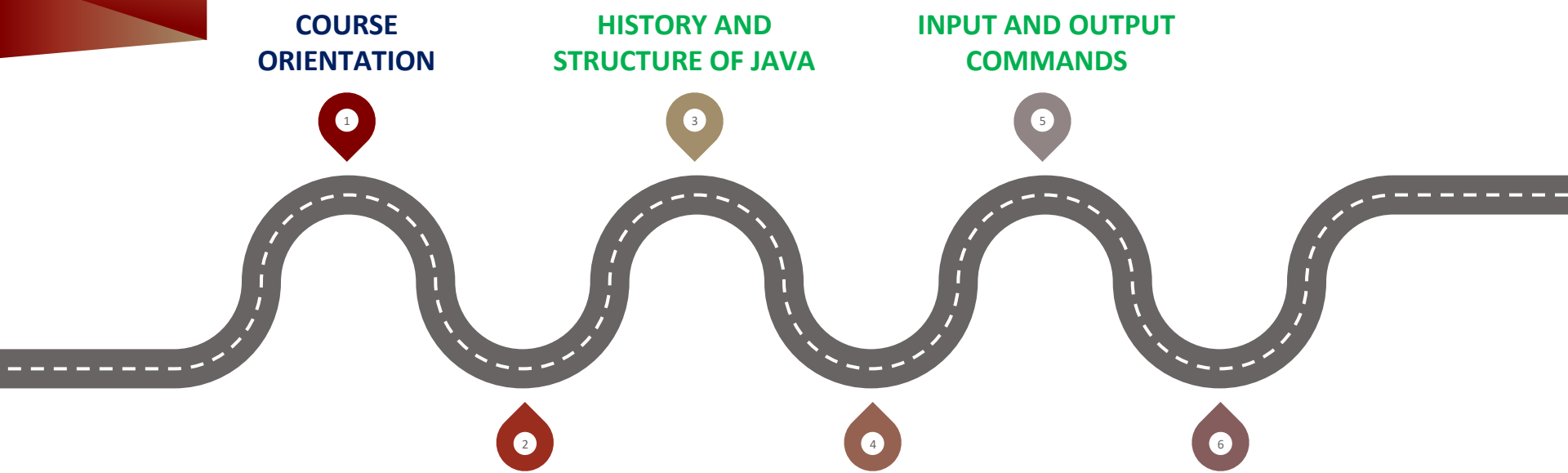


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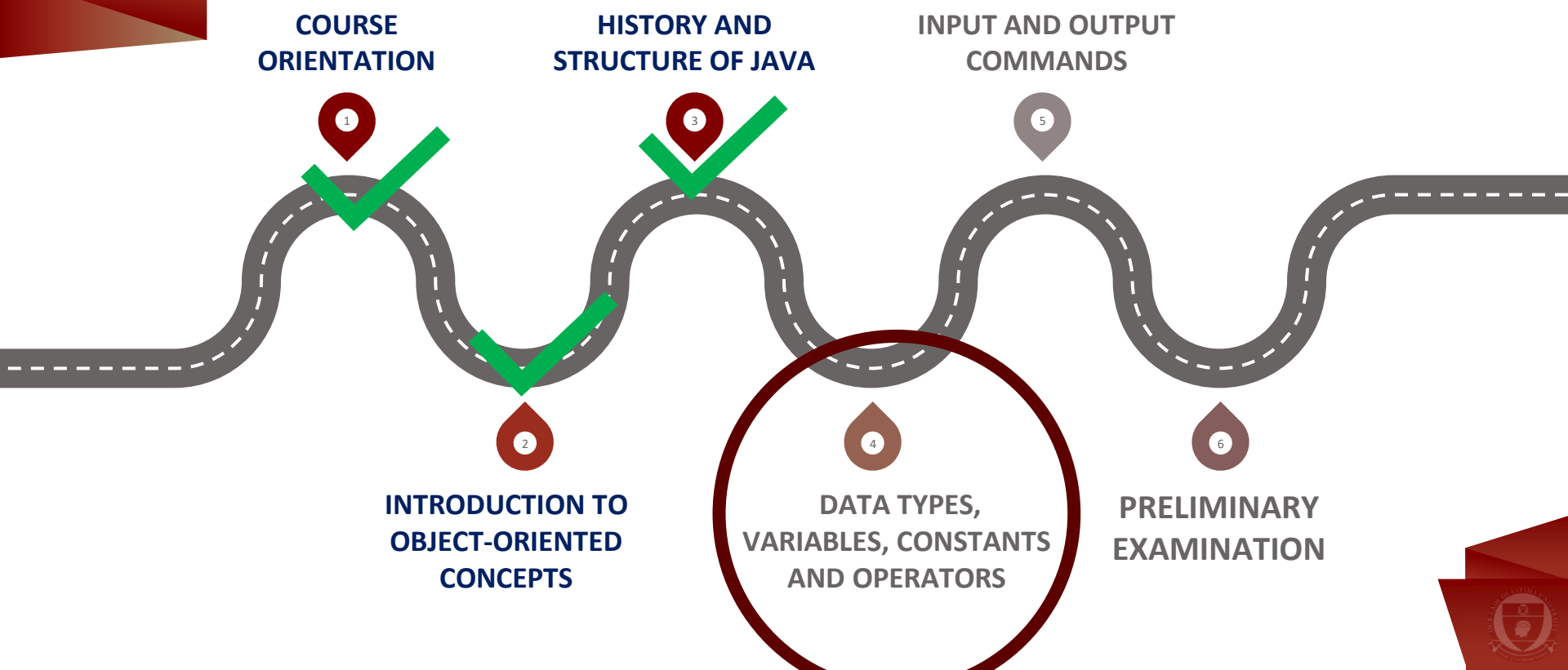


OOPR211
OBJECT-ORIENTED
PROGRAMMING I

PRELIMS ROADMAP



PRELIMS ROADMAP



Module 3: DATA TYPES, VARIABLES, CONSTANTS AND OPERATORS

Module Objectives:

1

Understand the concepts of different data types, variable declaration and operators.

2

Learn to use appropriate arithmetic, logical and relational operation strategies in different computing problems.

Topics:

- Java Data Types
- Declaring and Initializing Variables
- Constants
- Different Types of Operators in Java

1 Java Data Types

DATA TYPES

IN PROGRAMMING DATA TYPES IS ALWAYS INCLUDED IN VARIABLE DECLARATION.

A DATA TYPE IS ALWAYS WRITTEN BEFORE DEFINING THE VARIABLE NAME.

DATA TYPES

DATA TYPE CLASSIFIES A PARTICULAR TYPE OF INFORMATION AND DETERMINES WHAT KIND VALUE A VARIABLE COULD HOLD.

DATA TYPES

JAVA SUPPORTS TWO TYPES OF DATA TYPE:

THE PRIMITIVE DATA TYPE &

THE CLASS TYPE.

PRIMITIVE DATA TYPES

These are simple items or data types such as int, byte, long, short etc. Normally written in lower case.

8 PRIMITIVE DATA TYPES

Type Name	Kind of Value	Memory Used
byte	Integer	**1 byte**
short	Integer	**2 bytes**
int	Integer	**4 bytes**
long	Integer	**8 bytes**
float	Floating Point Numbers	**4 bytes**
double	Floating Point Numbers	**8 bytes**
char	Single Character	**2 bytes**
Boolean	**True or False**	**1 bit**

CLASS DATA TYPES

also known as non primitive data types. Type for object with both data and methods. Written in proper case.

CLASS DATA TYPES

Example:

String data type and any type created by a user.

2 Declaring and Initializing Variables

WHAT IS
VARIABLE



VARIABLE

IT IS A LOCATION IN THE COMPUTER'S MEMORY WHERE A VALUE CAN BE STORED FOR USE BY A PROGRAM.

VARIABLE

CAN HOLD ONLY ONE VALUE AT A TIME.

VARIABLE

VALUE CAN CHANGE.

SYNTAX FOR VARIABLE DECLARATION

Single

- [*DataType*] [*variable*];

Multiple

- [*DataType*] [*variable1*], [*variable2*];

EXAMPLES

- int number1;
- int number1, number2, sum;
- String name;
- float average;

Using Variables

```
public class UsingVariables_1
{
    public static void main(String args [])
        {
            String name, fname, sname;
            fname = "Lailanie";
            sname = "Calma";
            name = fname + " " + sname;
            System.out.println("Hello " + name + "!!!");
        }
}
```

Output:

Hello Lailanie Calma!!!

Using Variables

Example: Adding of two Numbers

```
public class UsingVariables_2
{
    public static void main(String args [])
        {
            int num1, num2, sum;
            num1 = 5;
            num2 = 6;
            sum = num1 + num2;
            System.out.println("The sum of " + num1 + " and " + num2 + " is equals to: " + sum);
        }
}
```

Output:

```
The sum of 5 and 6 is equals to: 11
```

3 Constants

WHAT IS A CONSTANT



CONSTANT

CONSTANTS ARE SIMILAR TO VARIABLES EXCEPT THAT THEY HOLD A **FIXED VALUE.**

THEY ARE ALSO CALLED “READ” ONLY VARIABLES.



CONSTANT

CANNOT BE CHANGED WHILE PROGRAM IS RUNNING.

CONSTANT

CONSTANTS ARE DECLARED WITH THE RESERVED WORD

“FINAL”.

CONSTANT

BY CONVENTION

UPPER CASE LETTERS

ARE USED FOR DEFINING CONSTANTS.

Using CONSTANT

```
public class CircleArea
{
   public static void main(String args[])
   {
        final double PI = 3.1428;
        double radius = 5.5; // in cms
        double area;

        area = PI * radius * radius;

        System.out.println("Circle Radius = "+radius+" Area="+area);
   }
}
```

4 Different Types of Operators in Java

OPERATORS

SYMBOLS USED TO DENOTE COMPUTATION, CONDITION OR RELATION ETC.

OPERATORS



OPERATORS ARE SYMBOLS WHICH HAS MEANING TO COMPUTER FOR IT DEFINES CONDITION, MATHEMATICAL OPERATIONS AS WELL AS LOGICAL OPERATIONS.

TYPES OF OPERATORS



Java™

Types of Operators



ARITHMETIC OPERATORS

MANIPULATE TWO OR MORE NUMERIC VALUES.

*** , / , + , -**

Types of Operators



COMPARISON OPERATORS

COMPARES
TWO NUMERIC VALUES OR CHARACTER VALUES.

> ,< ,<=, >= ,==, !=

Types of Operators

Math Notation	Name	Java Notation	Java Examples
=	Equal to	= =	Balance = = 0
=	Not equal	!=	answer != ‘y’
>	Greater than	>	expenses > income
>	Greater than or equal to	>=	Grade >=75
<	Less than	<	Grade < 75
<	Less than or equal to	<=	Grade<=75

Types of Operators



UNARY OPERATORS

THESE ARE USED FOR INCREMENTATION AND DECREMENTATION.

++, --

Types of Operators



ASSIGNMENT OPERATOR

NORMALLY SEEN IN A FORMULA LIKE STATEMENT KNOWN AS ASSIGNMENT STATEMENT.

=

Types of Operators



SPECIALIZED OPERATORS

IT COMBINES ASSIGNMENT AND ARITHMETIC OPERATORS.

***= , /= , += , -=**

Logical Operators

Operator	Symbol	Description
AND	&&	two or all of the given condition must be true in order for the result to be true.
OR	||	only one of the given condition must be true in order for the result to be true.
NOT	!	reverses the result of the given condition

QUESTIONS?

Thank you!

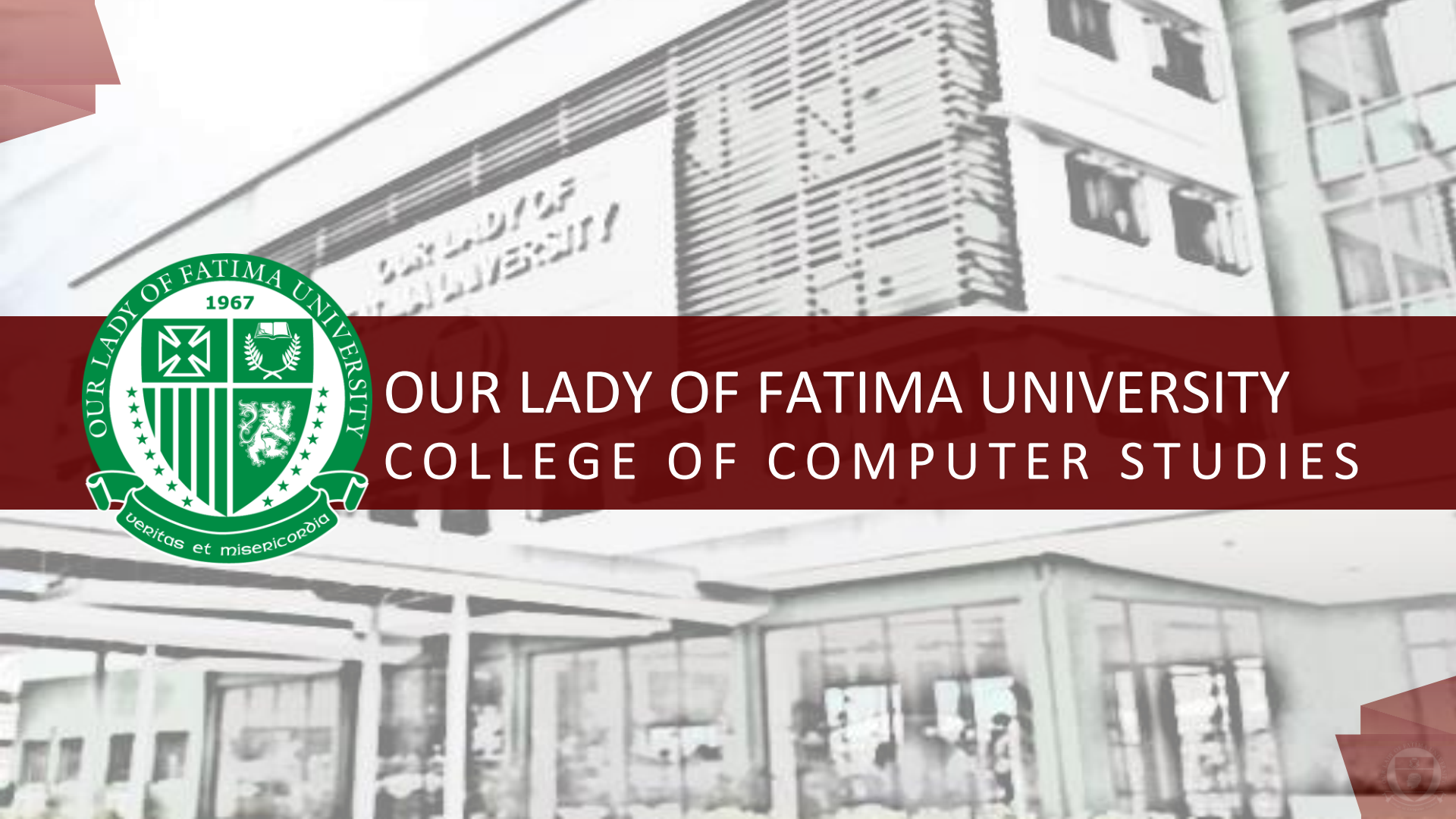
Credits

Special thanks to all the people who made and released these awesome resources for free:

- Presentation template by SlidesCarnival
- Photographs by Unsplash

Additional References:

- https://www.edureka.co/blog/what-is-java/
- http://ecomputernotes.com/java/what-is-java-language/types-of-java-programs



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